

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / co
Observed / expected	0.5 / 0.155725
Time	2026-07-10 16:00 UTC
Location	29.6697, -95.1285
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	7 / 724	43.23 to 100; mean 75.9835	66.44 at 2026-07-10 15:55Z; dt 5 min
asos	temperature	degC	7 / 724	23.6111 to 35.5556; mean 28.5678	31 at 2026-07-10 15:55Z; dt 5 min
asos	wind_direction	degree	7 / 896	0 to 360; mean 125.659	200 at 2026-07-10 15:55Z; dt 5 min
asos	wind_speed	m/s	7 / 918	0 to 10.8033; mean 2.5308	2.57222 at 2026-07-10 15:55Z; dt 5 min
noaa_gfs	gh_500	m	8 / 96	5905.85 to 5931.15; mean 5919.26	5920.27 at 2026-07-10 18:00Z; dt 120 min
noaa_gfs	pbl_height	m	8 / 96	12.0586 to 1950.8; mean 659.161	1686.16 at 2026-07-10 18:00Z; dt 120 min
noaa_gfs	precipitable_water	mm	8 / 96	43.0571 to 58.1093; mean 50.9795	55.9306 at 2026-07-10 18:00Z; dt 120 min
noaa_gfs	surface_pressure	hPa	8 / 96	1011.3 to 1017.46; mean 1014.15	1014.56 at 2026-07-10 18:00Z; dt 120 min
noaa_gfs	t_850	degC	8 / 96	17.2076 to 20.7322; mean 18.8377	17.2076 at 2026-07-10 18:00Z; dt 120 min
noaa_gfs	u_10m	m/s	8 / 96	-3.47027 to 1.76546; mean -0.7381	-2.37843 at 2026-07-10 18:00Z; dt 120 min
noaa_gfs	v_10m	m/s	8 / 96	-0.626672 to 5.52466; mean 2.283	2.66054 at 2026-07-10 18:00Z; dt 120 min
openaq	ozone	ppm	17 / 950	-0.003 to 0.054; mean 0.0164	0.023 at 2026-07-10 16:00Z; dt 0 min
openaq	pm10	ug/m3	2 / 119	9 to 60; mean 27.9664	24 at 2026-07-10 16:00Z; dt 0 min
openaq	pm25	ug/m3	77 / 3547	0 to 25.9; mean 5.3449	9.9 at 2026-07-10 16:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 360	0 to 100; mean 49.8444	79 at 2026-07-10 16:01Z; dt 2 min
openweather	humidity	percent	5 / 360	52 to 96; mean 78.6556	80 at 2026-07-10 16:01Z; dt 2 min
openweather	precipitation	mm/h	5 / 360	0 to 7.7; mean 0.0798	0 at 2026-07-10 16:01Z; dt 2 min
openweather	pressure	hPa	5 / 360	1014 to 1018; mean 1016.06	1017 at 2026-07-10 16:01Z; dt 2 min
openweather	temperature	degC	5 / 360	25.01 to 35.63; mean 29.0433	30.26 at 2026-07-10 16:01Z; dt 2 min
openweather	wind_direction	degree	5 / 360	0 to 333; mean 165.558	161 at 2026-07-10 16:01Z; dt 2 min
openweather	wind_speed	m/s	5 / 360	0.45 to 6.21; mean 2.3174	3.41 at 2026-07-10 16:01Z; dt 2 min
purpleair	pm25	ug/m3	65 / 4458	0 to 102.8; mean 6.8215	7.7 at 2026-07-10 15:59Z; dt 0.1 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-10 19:11Z; dt 191.4 min
sentinel5p	s5p_ch4_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-07-10 19:11Z; dt 191.4 min
sentinel5p	s5p_co_column	mol/m^2	4 / 4	0.0242691 to 0.0262775; mean 0.0253	0.0262775 at 2026-07-10 19:11Z; dt 191.4 min
sentinel5p	s5p_co_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-10 19:11Z; dt 191.4 min
sentinel5p	s5p_hcho_column	mol/m^2	4 / 4	0.000148478 to 0.000160047; mean 0.0002	0.000160047 at 2026-07-10 19:11Z; dt 191.4 min
sentinel5p	s5p_hcho_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-10 19:11Z; dt 191.4 min
sentinel5p	s5p_no2_column	mol/m^2	2 / 2	3.3321e-05 to 4.40738e-05; mean 0	4.40738e-05 at 2026-07-10 19:39Z; dt 219.1 min
sentinel5p	s5p_no2_granule_available	count	3 / 3	1 to 1; mean 1	1 at 2026-07-10 19:39Z; dt 219.1 min
sentinel5p	s5p_o3_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-10 19:11Z; dt 191.4 min
sentinel5p	s5p_so2_column	mol/m^2	4 / 4	-5.8828e-06 to 2.37917e-05; mean 0	2.37917e-05 at 2026-07-10 19:11Z; dt 191.4 min
sentinel5p	s5p_so2_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-10 19:11Z; dt 191.4 min
tceq	co	ppm	3 / 191	0.1 to 0.9; mean 0.2445	0.5 at 2026-07-10 16:00Z; dt 0 min
tceq	no2	ppb	12 / 745	-1.9 to 20.3; mean 5.4154	1.9 at 2026-07-10 16:00Z; dt 0 min
tceq	so2	ppb	3 / 185	-0.6 to 3.7; mean -0.0978	0.2 at 2026-07-10 16:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	07-09 00Z 82.72, 06Z 88.61, 12Z 67.82, 18Z 56.47, 07-10 00Z 75.84, 06Z 87.02, 12Z 74.65, 18Z 62.43, 07-11 00Z 78.3, 06Z 88.71, 12Z 79.97, 18Z 70.7, 07-12 00Z 85.94
asos	temperature	07-09 00Z 26.99, 06Z 25.75, 12Z 30.42, 18Z 32.36, 07-10 00Z 28.37, 06Z 26.29, 12Z 29.18, 18Z 31.39, 07-11 00Z 27.77, 06Z 26.07, 12Z 27.89, 18Z 29.44, 07-12 00Z 27.19
asos	wind_direction	07-09 00Z 143.5, 06Z 121.7, 12Z 163.3, 18Z 171.7, 07-10 00Z 146.1, 06Z 80.67, 12Z 118.7, 18Z 156.7, 07-11 00Z 135.7, 06Z 64.13, 12Z 131.6, 18Z 108.1, 07-12 00Z 91.6
asos	wind_speed	07-09 00Z 2.196, 06Z 1.418, 12Z 2.376, 18Z 5.23, 07-10 00Z 2.779, 06Z 1.128, 12Z 2.412, 18Z 4.476, 07-11 00Z 2.281, 06Z 0.8867, 12Z 2.258, 18Z 2.762, 07-12 00Z 2.315
noaa_gfs	gh_500	07-09 06Z 5919, 12Z 5910, 18Z 5921, 07-10 00Z 5915, 06Z 5921, 12Z 5907, 18Z 5920, 07-11 00Z 5919, 06Z 5924, 12Z 5919, 18Z 5930, 07-12 00Z 5927
noaa_gfs	pbl_height	07-09 06Z 291.9, 12Z 144.3, 18Z 1731, 07-10 00Z 883.8, 06Z 203.7, 12Z 115.4, 18Z 1552, 07-11 00Z 975.9, 06Z 280.3, 12Z 70.06, 18Z 854.7, 07-12 00Z 807.6
noaa_gfs	precipitable_water	07-09 06Z 44.47, 12Z 44.38, 18Z 45.86, 07-10 00Z 46.68, 06Z 46.47, 12Z 53.7, 18Z 55.67, 07-11 00Z 53.48, 06Z 52.82, 12Z 55.97, 18Z 55.83, 07-12 00Z 56.43
noaa_gfs	surface_pressure	07-09 06Z 1015, 12Z 1015, 18Z 1015, 07-10 00Z 1013, 06Z 1013, 12Z 1014, 18Z 1014, 07-11 00Z 1013, 06Z 1014, 12Z 1015, 18Z 1016, 07-12 00Z 1014
noaa_gfs	t_850	07-09 06Z 19.44, 12Z 19.05, 18Z 17.77, 07-10 00Z 18.86, 06Z 20.5, 12Z 18.53, 18Z 17.67, 07-11 00Z 18.52, 06Z 19.37, 12Z 18.76, 18Z 18.34, 07-12 00Z 19.26
noaa_gfs	u_10m	07-09 06Z 0.9761, 12Z 1.052, 18Z -0.326, 07-10 00Z -0.9792, 06Z -0.937, 12Z 0.1127, 18Z -2.221, 07-11 00Z -1.981, 06Z -1.098, 12Z -0.3907, 18Z -0.4401, 07-12 00Z -2.624

Source	Metric	Values
noaa_gfs	v_10m	07-09 06Z 2.668, 12Z 1.384, 18Z 2.518, 07-10 00Z 4.73, 06Z 2.002, 12Z 1.276, 18Z 3.214, 07-11 00Z 3.62, 06Z 2.315, 12Z 0.5108, 18Z 1.001, 07-12 00Z 2.157
openaq	ozone	07-09 12Z 0.01656, 18Z 0.0252, 07-10 00Z 0.01537, 06Z 0.006988, 12Z 0.01161, 18Z 0.02268, 07-11 00Z 0.01203, 06Z 0.006133, 12Z 0.01238, 18Z 0.02849, 07-12 00Z 0.01893
openaq	pm10	07-09 12Z 27.25, 18Z 26.75, 07-10 00Z 15.5, 06Z 18.83, 12Z 28.33, 18Z 35.27, 07-11 00Z 42.45, 06Z 33.08, 12Z 34.27, 18Z 23.17, 07-12 00Z 23.89
openaq	pm25	07-09 00Z 4.437, 06Z 4.489, 12Z 4.986, 18Z 4.063, 07-10 00Z 3.709, 06Z 5.629, 12Z 6.406, 18Z 5.188, 07-11 00Z 6.032, 06Z 6.073, 12Z 5.549, 18Z 5.589, 07-12 00Z 5.505
openweather	cloud_cover	07-09 00Z 49.2, 06Z 31.8, 12Z 3.733, 18Z 23.33, 07-10 00Z 88.62, 06Z 47.74, 12Z 84.67, 18Z 87.21, 07-11 00Z 25, 06Z 11.03, 12Z 73.84, 18Z 88.41, 07-12 00Z 29.48
openweather	humidity	07-09 00Z 79.9, 06Z 88.2, 12Z 76.87, 18Z 60.7, 07-10 00Z 74.24, 06Z 88.13, 12Z 79.8, 18Z 66.86, 07-11 00Z 76.9, 06Z 87.79, 12Z 86.58, 18Z 74, 07-12 00Z 84.1
openweather	precipitation	07-09 00Z 0, 06Z 0, 12Z 0, 18Z 0.2567, 07-10 00Z 0, 06Z 0, 12Z 0.1113, 18Z 0.003448, 07-11 00Z 0, 06Z 0, 12Z 0.3806, 18Z 0.1293, 07-12 00Z 0.09762
openweather	pressure	07-09 00Z 1017, 06Z 1016, 12Z 1018, 18Z 1015, 07-10 00Z 1015, 06Z 1015, 12Z 1017, 18Z 1015, 07-11 00Z 1015, 06Z 1016, 12Z 1018, 18Z 1016, 07-12 00Z 1016
openweather	temperature	07-09 00Z 28.01, 06Z 26.28, 12Z 29.95, 18Z 34.18, 07-10 00Z 29.64, 06Z 26.49, 12Z 29.45, 18Z 32.61, 07-11 00Z 28.81, 06Z 26.15, 12Z 27.24, 18Z 30.08, 07-12 00Z 27.85
openweather	wind_direction	07-09 00Z 160.8, 06Z 172.6, 12Z 216.1, 18Z 179.6, 07-10 00Z 182.1, 06Z 171.2, 12Z 171.4, 18Z 150.1, 07-11 00Z 146, 06Z 142.9, 12Z 162.5, 18Z 145, 07-12 00Z 140.6
openweather	wind_speed	07-09 00Z 2.706, 06Z 1.932, 12Z 1.636, 18Z 2.793, 07-10 00Z 2.224, 06Z 1.664, 12Z 1.934, 18Z 3.691, 07-11 00Z 1.956, 06Z 1.898, 12Z 2.659, 18Z 3.01, 07-12 00Z 2.373
purpleair	pm25	07-09 00Z 5.683, 06Z 6.844, 12Z 6.313, 18Z 3.945, 07-10 00Z 5.904, 06Z 7.399, 12Z 7.635, 18Z 6.472, 07-11 00Z 8.086, 06Z 8.661, 12Z 7.124, 18Z 6.638, 07-12 00Z 7.495
sentinel5p	s5p_aer_ai_granule_available	07-09 18Z 1, 07-10 18Z 1, 07-11 18Z 1
sentinel5p	s5p_ch4_granule_available	07-09 18Z 1, 07-10 18Z 1, 07-11 18Z 1
sentinel5p	s5p_co_column	07-09 18Z 0.02438, 07-10 18Z 0.02624
sentinel5p	s5p_co_granule_available	07-09 18Z 1, 07-10 18Z 1, 07-11 18Z 1
sentinel5p	s5p_hcho_column	07-09 18Z 0.0001505, 07-10 18Z 0.0001571
sentinel5p	s5p_hcho_granule_available	07-09 18Z 1, 07-10 18Z 1, 07-11 18Z 1
sentinel5p	s5p_no2_column	07-09 18Z 3.332e-05, 07-10 18Z 4.407e-05
sentinel5p	s5p_no2_granule_available	07-09 18Z 1, 07-10 18Z 1, 07-11 18Z 1
sentinel5p	s5p_o3_granule_available	07-09 18Z 1, 07-10 18Z 1, 07-11 18Z 1
sentinel5p	s5p_so2_column	07-09 18Z 2.306e-05, 07-10 18Z 8.954e-06
sentinel5p	s5p_so2_granule_available	07-09 18Z 1, 07-10 18Z 1, 07-11 18Z 1
tceq	co	07-09 06Z 0.2222, 12Z 0.2722, 18Z 0.1667, 07-10 00Z 0.1667, 06Z 0.2167, 12Z 0.2722, 18Z 0.2278, 07-11 00Z 0.21, 06Z 0.2389, 12Z 0.3167, 18Z 0.2389, 07-12 00Z 0.4
tceq	no2	07-09 06Z 5.834, 12Z 5.389, 18Z 3.253, 07-10 00Z 2.408, 06Z 5.294, 12Z 6.969, 18Z 5.342, 07-11 00Z 4.884, 06Z 6.561, 12Z 5.349, 18Z 5.765, 07-12 00Z 6.958
tceq	so2	07-09 06Z -0.1556, 12Z -0.09412, 18Z -0.1667, 07-10 00Z -0.1444, 06Z -0.2056, 12Z -0.1833, 18Z -0.1889, 07-11 00Z -0.12, 06Z -0.1944, 12Z -0.1833, 18Z 0.5389, 07-12 00Z -0.14

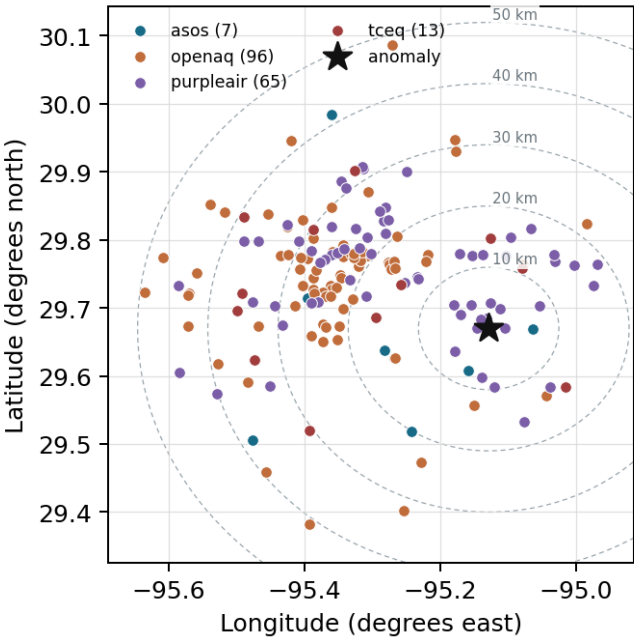
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

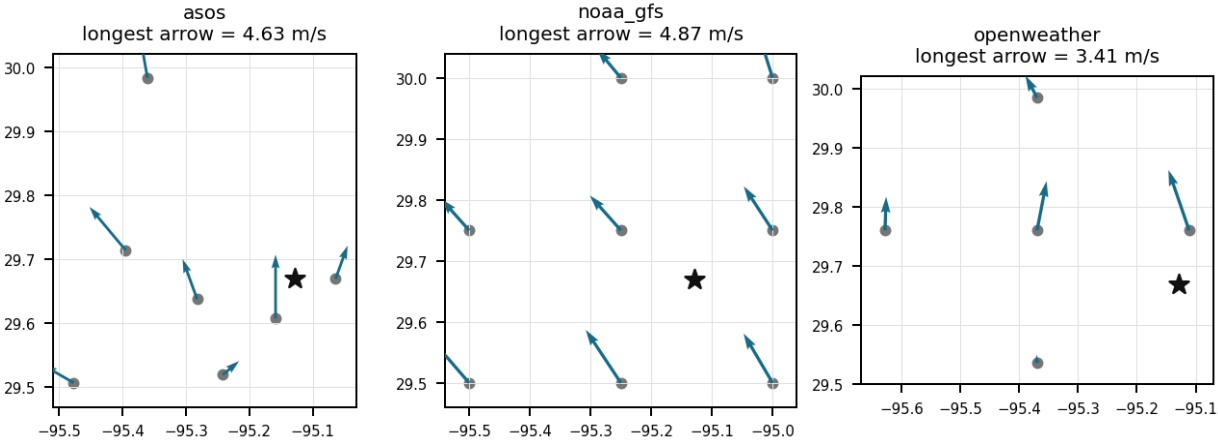
Source	Metric	Values
asos	humidity	T41 (6.225 km) 77.2, EFD (7.53 km) 74.79, HOU (15.295 km) 75.57, LVJ (20.026 km) 74.49, MCJ (26.207 km) 75.26, AXH (38.286 km) 76.84, IAH (41.543 km) 76.37
asos	temperature	T41 (6.225 km) 28.77, EFD (7.53 km) 29, HOU (15.295 km) 28.74, LVJ (20.026 km) 28.8, MCJ (26.207 km) 28.7, AXH (38.286 km) 27.72, IAH (41.543 km) 28.78
asos	wind_direction	T41 (6.225 km) 148.9, EFD (7.53 km) 125.8, HOU (15.295 km) 144.6, LVJ (20.026 km) 100.6, MCJ (26.207 km) 151.1, AXH (38.286 km) 75.24, IAH (41.543 km) 137.4
asos	wind_speed	T41 (6.225 km) 2.822, EFD (7.53 km) 3.132, HOU (15.295 km) 2.879, LVJ (20.026 km) 1.901, MCJ (26.207 km) 3.655, AXH (38.286 km) 1.043, IAH (41.543 km) 2.651
noaa_gfs	gh_500	gfs:29.75,-95.25 (14.741 km) 5919, gfs:29.75,-95.00 (15.288 km) 5920, gfs:29.50,-95.25 (22.231 km) 5919, gfs:29.50,-95.00 (22.598 km) 5920, gfs:29.75,-95.50 (36.97 km) 5919, gfs:30.00,-95.25 (38.548 km) 5919, gfs:30.00,-95.00 (38.76 km) 5920, gfs:29.50,-95.50 (40.577 km) 5919
noaa_gfs	pbl_height	gfs:29.75,-95.25 (14.741 km) 762, gfs:29.75,-95.00 (15.288 km) 727.3, gfs:29.50,-95.25 (22.231 km) 457, gfs:29.50,-95.00 (22.598 km) 585.9, gfs:29.75,-95.50 (36.97 km) 792, gfs:30.00,-95.25 (38.548 km) 838.9, gfs:30.00,-95.00 (38.76 km) 545.4, gfs:29.50,-95.50 (40.577 km) 564.7
noaa_gfs	precipitable_water	gfs:29.75,-95.25 (14.741 km) 50.77, gfs:29.75,-95.00 (15.288 km) 51.12, gfs:29.50,-95.25 (22.231 km) 51.45, gfs:29.50,-95.00 (22.598 km) 51.83, gfs:29.75,-95.50 (36.97 km) 50.74, gfs:30.00,-95.25 (38.548 km) 50.32, gfs:30.00,-95.00 (38.76 km) 50.55, gfs:29.50,-95.50 (40.577 km) 51.04
noaa_gfs	surface_pressure	gfs:29.75,-95.25 (14.741 km) 1014, gfs:29.75,-95.00 (15.288 km) 1015, gfs:29.50,-95.25 (22.231 km) 1015, gfs:29.50,-95.00 (22.598 km) 1016, gfs:29.75,-95.50 (36.97 km) 1013, gfs:30.00,-95.25 (38.548 km) 1013, gfs:30.00,-95.00 (38.76 km) 1014, gfs:29.50,-95.50 (40.577 km) 1014
noaa_gfs	t_850	gfs:29.75,-95.25 (14.741 km) 18.82, gfs:29.75,-95.00 (15.288 km) 18.87, gfs:29.50,-95.25 (22.231 km) 18.82, gfs:29.50,-95.00 (22.598 km) 18.86, gfs:29.75,-95.50 (36.97 km) 18.75, gfs:30.00,-95.25 (38.548 km) 18.86, gfs:30.00,-95.00 (38.76 km) 18.88, gfs:29.50,-95.50 (40.577 km) 18.83
noaa_gfs	u_10m	gfs:29.75,-95.25 (14.741 km) -0.4465, gfs:29.75,-95.00 (15.288 km) -0.6573, gfs:29.50,-95.25 (22.231 km) -0.7732, gfs:29.50,-95.00 (22.598 km) -0.6898, gfs:29.75,-95.50 (36.97 km) -0.8248, gfs:30.00,-95.25 (38.548 km) -0.6923, gfs:30.00,-95.00 (38.76 km) -0.7207, gfs:29.50,-95.50 (40.577 km) -1.1
noaa_gfs	v_10m	gfs:29.75,-95.25 (14.741 km) 2.325, gfs:29.75,-95.00 (15.288 km) 2.295, gfs:29.50,-95.25 (22.231 km) 2.213, gfs:29.50,-95.00 (22.598 km) 2.727, gfs:29.75,-95.50 (36.97 km) 2.317, gfs:30.00,-95.25 (38.548 km) 2.033, gfs:30.00,-95.00 (38.76 km) 1.957, gfs:29.50,-95.50 (40.577 km) 2.397
openaq	ozone	332 (0.032 km) 0.021, 2800 (11.598 km) 0.01768, 2039 (14.096 km) 0.01569, 320 (14.272 km) 0.01437, 3378 (14.356 km) 0.01327, 5077791 (14.558 km) 0.01547, 3214 (14.765 km) 0.01603, 325 (16.163 km) 0.004614, +9 more
openaq	pm10	271 (0.032 km) 23.33, 15852419 (21.358 km) 32.68
openaq	pm25	268 (0.032 km) 9.193, 13955815 (12.727 km) 5.355, 14404954 (13.725 km) 2.613, 14157975 (14.275 km) 8.603, 15539682 (14.359 km) 6.969, 8967123 (14.413 km) 7.731, 8967479 (14.413 km) 7.161, 13787195 (14.82 km) 7.406, +69 more
openweather	cloud_cover	grid:east (10.221 km) 51.78, grid:center (25.388 km) 48.85, grid:south (27.676 km) 51.62, grid:north (42.079 km) 51.93, grid:west (49.325 km) 45.04
openweather	humidity	grid:east (10.221 km) 81.99, grid:center (25.388 km) 78.29, grid:south (27.676 km) 77.64, grid:north (42.079 km) 77.51, grid:west (49.325 km) 77.85
openweather	precipitation	grid:east (10.221 km) 0.1008, grid:center (25.388 km) 0.08833, grid:south (27.676 km) 0.07028, grid:north (42.079 km) 0.1371, grid:west (49.325 km) 0.002639
openweather	pressure	grid:east (10.221 km) 1016, grid:center (25.388 km) 1016, grid:south (27.676 km) 1016, grid:north (42.079 km) 1016, grid:west (49.325 km) 1016
openweather	temperature	grid:east (10.221 km) 28.94, grid:center (25.388 km) 29.33, grid:south (27.676 km) 28.78, grid:north (42.079 km) 29.16, grid:west (49.325 km) 29.01
openweather	wind_direction	grid:east (10.221 km) 176.6, grid:center (25.388 km) 166.8, grid:south (27.676 km) 171.5, grid:north (42.079 km) 138.1, grid:west (49.325 km) 174.8
openweather	wind_speed	grid:east (10.221 km) 3.105, grid:center (25.388 km) 1.992, grid:south (27.676 km) 2.635, grid:north (42.079 km) 1.882, grid:west (49.325 km) 1.972
purpleair	pm25	128007 (1.625 km) 1.195, 161689 (1.727 km) 5.793, 222589 (1.859 km) 6.249, 268165 (2.296 km) 6.965, 222593 (3.655 km) 6.504, 250205 (4.119 km) 4.371, 98633 (4.643 km) 1.442, 244939 (4.684 km) 5.954, +57 more
tceq	co	482011039 (0.0 km) 0.1609, 482011035 (14.355 km) 0.1365, 482011052 (29.756 km) 0.4344

Source	Metric	Values
tceq	no2	482011039 (0.0 km) 3.528, 482011015 (10.998 km) 4.617, 482011035 (14.355 km) 8.44, 482011050 (14.566 km) 1.461, 482010026 (14.787 km) 5.9, 482010416 (16.162 km) 5.108, 482011052 (29.756 km) 12.11, 480391004 (30.447 km) 3.469, +4 more
tceq	so2	482011039 (0.0 km) 0.2806, 482011035 (14.355 km) -0.2016, 482010051 (33.805 km) -0.3742

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

A short-duration local combustion plume from nearby industrial equipment, flaring, ships, rail, trucks, or other fuel-burning sources is a plausible cause of the CO spike because the anomaly was large at one TCEQ monitor while nearby NO₂ and SO₂ at the same time remained low, a pattern consistent with a narrow CO-rich plume rather than a broad regional photochemical event.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 2 of 36

The TCEQ carbon monoxide anomaly at 29.670, -95.129 had a z-score of 5.91885943430829 and was classified as severe.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 3 of 36

The localized ground-level impact at TCEQ monitor 482011039 is supported by substantially lower contemporaneous carbon monoxide levels recorded at nearby TCEQ monitor 482011035.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 4 of 36

A shallow morning boundary layer and light surface winds restricted atmospheric dispersion, trapping surface carbon monoxide emissions near the ground prior to afternoon boundary layer growth.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 5 of 36

The air quality anomaly in Houston, Texas on July 10, 2026 was likely caused by a combination of factors including weather patterns and local emissions.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 6 of 36

The temperature was unusually high on July 10th, 2026, which may have contributed to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 7 of 36

Ground monitor 482011039 recorded a peak carbon monoxide concentration of 0.5 ppm at 16:00 UTC on July 10, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 8 of 36

Satellite observations from Sentinel-5P detected an elevated total column carbon monoxide density over the broader Houston area on July 10, 2026, supporting a regional enhancement in carbon monoxide.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 9 of 36

A severe carbon monoxide concentration anomaly was detected at location (29.670, -95.129) on 2026-07-10T16:00:00+00:00 with an observed value of 0.5 ppm compared to an expected mean of approximately 0.156 ppm, resulting in a z-score of 5.92.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 10 of 36

A strong wind with a speed of over 3.41 m/s blew through Houston on July 10th, 2026, potentially dispersing pollutants and affecting air quality.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 11 of 36

Regional smoke or area-wide secondary pollution is a less plausible primary cause of the CO spike because PM2.5 and PM10 near the event were only modestly elevated, ozone was not unusually high, and the satellite CO column enhancement was only slight relative to the prior overpass day.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 12 of 36

The air quality anomaly is related to particulate matter (PM2.5) with a value of 7.7 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 13 of 36

The anomalous pollutant was carbon monoxide measured by a TCEQ monitor at 29.670, -95.129 in Houston at 2026-07-10T16:00:00+00:00.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 14 of 36

The air quality anomaly in Houston, Texas is caused by high levels of SO2 particles.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 15 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 16 of 36

Meteorological transport from sources to the south or southwest of the monitor is a plausible cause of the CO spike because surface winds near the event were from about 161 to 200 degrees with light-to-moderate speeds near 2.6 to 3.4 m/s, which would advect emissions from the Gulf/Ship Channel side toward the site while still allowing elevated near-source concentrations.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 17 of 36

ASOS weather observations near the anomaly recorded surface wind speeds of approximately 2.5 m/s from the south-southeast around 16:00 UTC on July 10, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 18 of 36

Surface winds near the anomaly were southerly to southwesterly at roughly 2.6 to 3.4 m/s around 16:00 UTC, which is consistent with transport of a nearby combustion plume into the monitor rather than calm stagnation alone.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 19 of 36

The air quality anomaly in Houston, Texas on July 10, 2026 was also characterized by elevated levels of NO₂.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 20 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO₂ particles.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 21 of 36

Sentinel-5P atmospheric observations recorded an increase in total column carbon monoxide over the Houston region from 0.0244 mol/m² on July 9, 2026, to 0.0263 mol/m² on July 10, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 22 of 36

The TCEQ carbon monoxide concentration at 29.670, -95.129 was 0.5 ppm at 2026-07-10T16:00:00+00:00, compared with an expected value of 0.15572519083969463 ppm.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 23 of 36

The wind direction at the time of the anomaly was approximately 161.0 degrees.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 24 of 36

Sentinel-5P satellite measurements support a broader regional enhancement, recording an increase in total column carbon monoxide from 0.0244 mol/m² on July 9 to 0.0262 mol/m² on July 10.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 25 of 36

The regional-stagnation-or-biomass-smoke hypothesis is contradicted because boundary-layer depth was high by afternoon near 1686 m rather than strongly suppressed, PM2.5 near the site was only about 7.7 to 9.9 ug/m3, PM10 near the site was 24 ug/m3, ozone near the site was 0.023 ppm, and satellite SO2 was not elevated on the same day, so the data do not show a broad smoke or multi-pollutant accumulation episode accompanying the CO spike.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 26 of 36

Industrial combustion or process emissions originating from facilities upwind in the Houston Ship Channel region contributed directly to the elevated surface carbon monoxide concentration recorded at TCEQ monitor 482011039.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 27 of 36

The air quality anomaly in Houston, Texas on July 10, 2026 was characterized by elevated levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 28 of 36

Co-pollutants and particles near the anomaly were not elevated in a way expected for a broad regional smoke or photochemical event, because the coincident site values were NO2 1.9 ppb and SO2 0.2 ppb from tceq, ozone 0.023 ppm from openaq, PM2.5 9.9 ug/m3 from openaq, and PM2.5 7.7 ug/m3 from purpleair.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 29 of 36

The TCEQ monitor at 29.670, -95.129 measured CO of 0.5 ppm at 2026-07-10 16:00 UTC, far above the site-specific expected value of 0.1557 ppm, indicating a severe localized CO enhancement.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 30 of 36

TCEQ ground monitor 482011039, positioned at the anomaly location (29.670, -95.129), recorded the elevated carbon monoxide reading of 0.5 ppm at 16:00 UTC on July 10, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 31 of 36

NOAA GFS boundary layer height data supports atmospheric trapping during the morning hours, showing a low mixing depth of 115.4 meters around 12:00 UTC on July 10 before expanding to 1552 meters by 18:00 UTC.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 32 of 36

The PM2.5 levels were higher than usual in the area, possibly due to nearby industrial or agricultural activities, which could have contributed to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 33 of 36

Satellite measurements from Sentinel-5P showed higher total column carbon monoxide levels over the Houston region on July 10, 2026, than on July 9, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 34 of 36

The local-combustion-plume hypothesis is also supported because co-pollutants associated with fresh industrial or mobile combustion were muted at the anomaly time, with site NO₂ at 1.9 ppb and site SO₂ at 0.2 ppb, which is consistent with a CO-enhanced plume from a source or plume age that does not strongly elevate NO₂ or SO₂ at the receptor.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 35 of 36

The local-combustion-plume hypothesis is supported because the TCEQ monitor at the anomaly site recorded CO of 0.5 ppm against a site mean near 0.1609 ppm, while winds near the event were from roughly 161 to 200 degrees at about 2.6 to 3.4 m/s, a pattern consistent with transport from sources located south to south-southwest of the monitor.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 36 of 36

The temperature at the time of the anomaly was approximately 30.26°C.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
